

Evaluation: Potential CLL-Classifiers

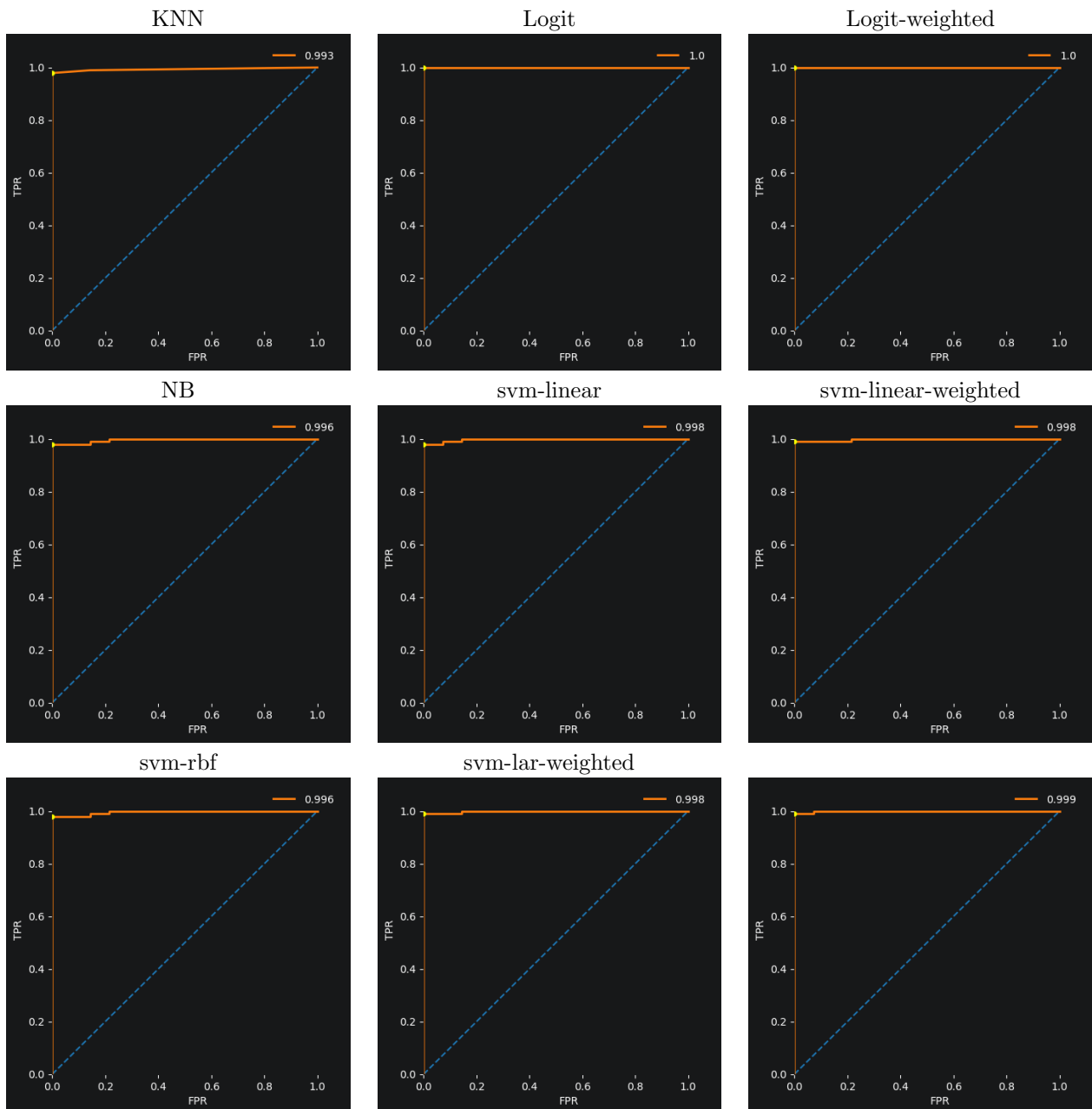
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1 Exp 01: Full Batch RMA

Since I introduced here intentionally strong data leakage, I will not investigate further after inspecting ROC curves of the trained model onto GSE131599-test.

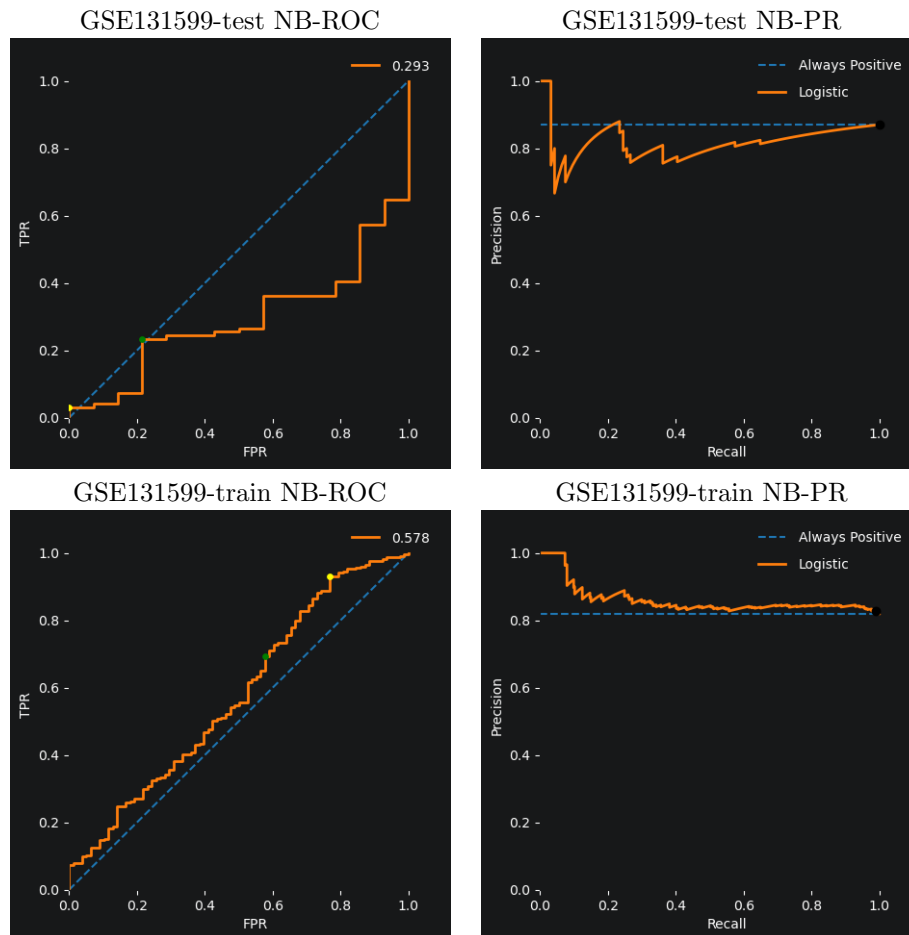
GSE131599-test (ROC)



- I intentionally produce some data leakage, I wanted to see the effect for scaling experiments combined with RMA
- This approach is clearly not valid, since it produces over-promising results
- Big Red Flag !
- Furthermore, also the prediction of GSE16455-test yields near perfect results, but i will not add those plots this report
- I will not further investigate here

2 Exp 02: Standard Scaling

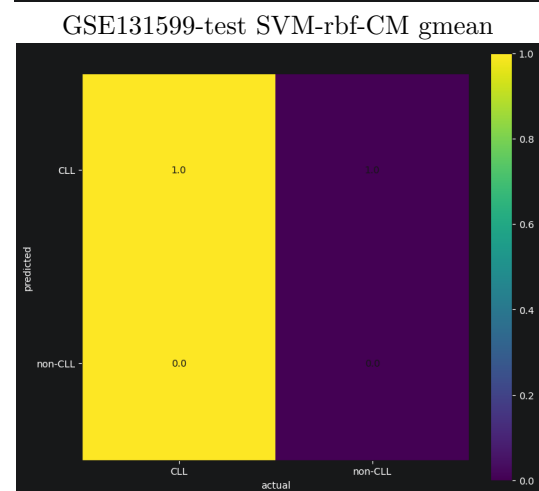
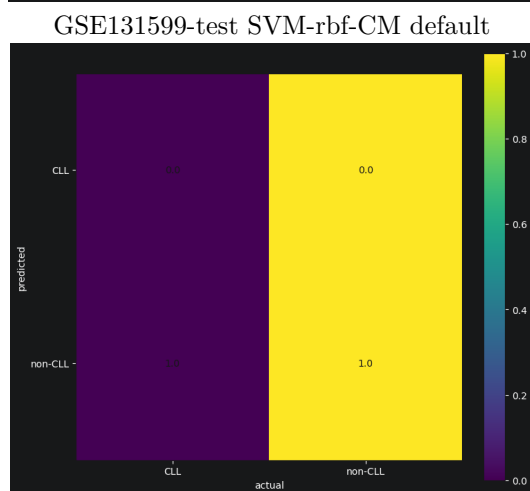
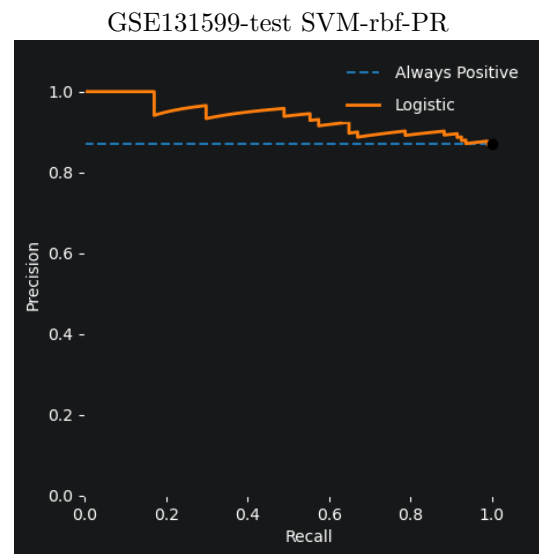
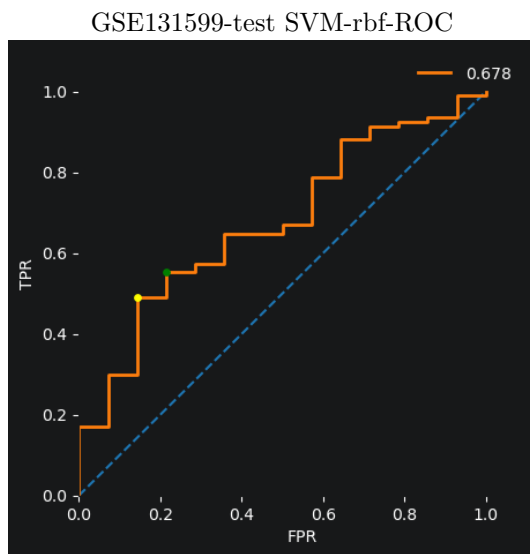
Most of the models ROC-AUC of the GSE131599-test data set was mostly around 0.5 and is therefore not a valid model for a diagnostic classifier. I only found that the Naive Bayes model reached a ROC-AUC of 0.293. I investigated further, since one could flip the prediction logic here, to get a ROC-AUC of roughly 0.7.



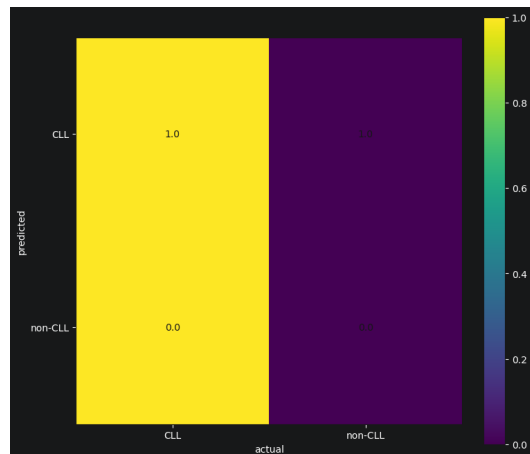
- Flipping the prediction logic, would be cherry-picking on the test data
- Also, it would make the train evaluation badly ...
- Conclusion, Standard Scaling with a Naive Bayes doesn't yield a promising classifier

3 Exp 03: Robust Scaling

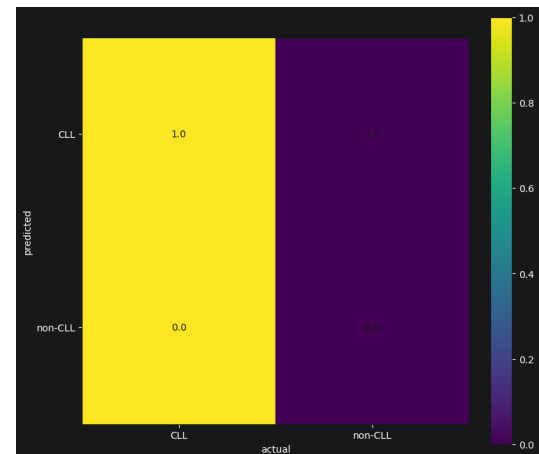
I found that the SVM-rbf performed onto the GSE13159-test very reasonable, so I investigated its performance further. I included the inspection of the confusion matrix (CM), when using different optimal threshold evaluated onto the train.



GSE131599-test SVM-rbf-CM Youden's J statistic



GSE131599-test SVM-rbf-CM F1-Score

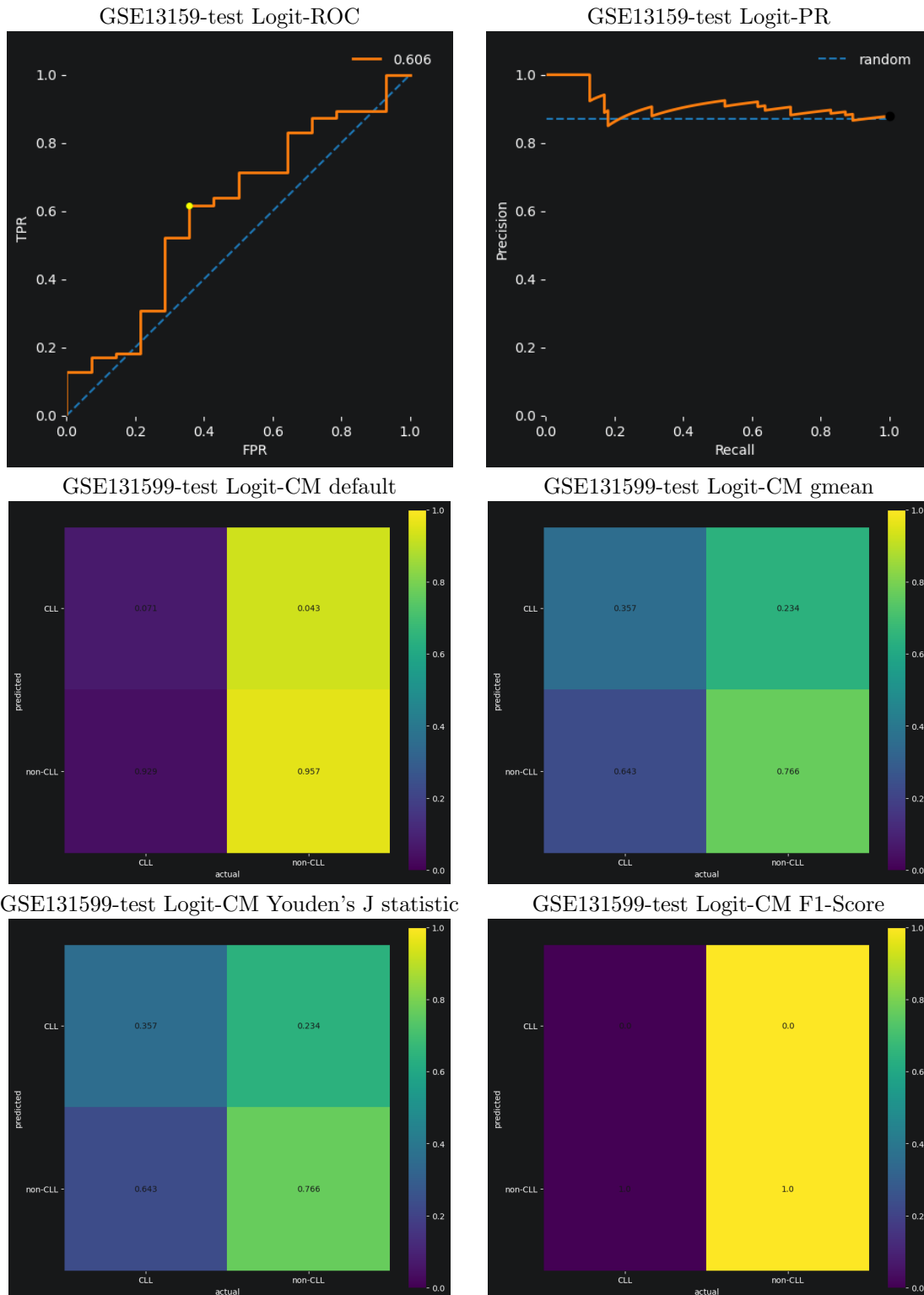


- One can clearly reason, that the ROC and PRC are promising, but looking at the actual predictions, the model only outputs a single class. that's a big Red flag, since the model doesn't generalize well (even with default threshold)

4 Exp 04: RMA-train with Approximation-test

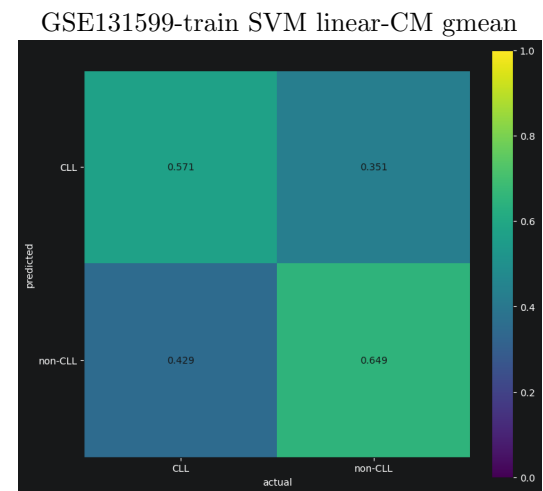
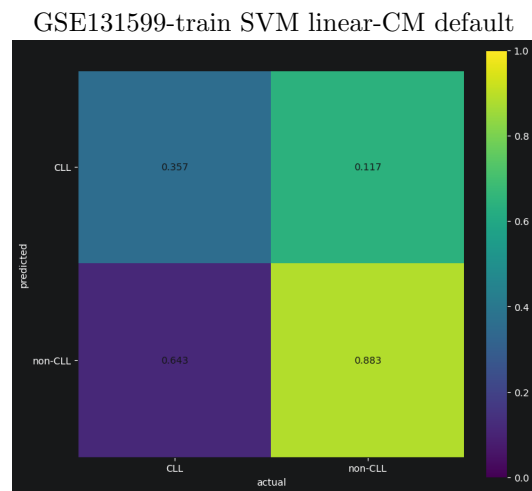
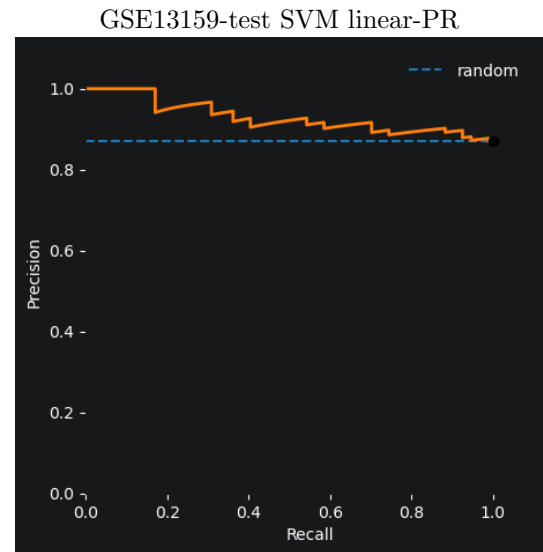
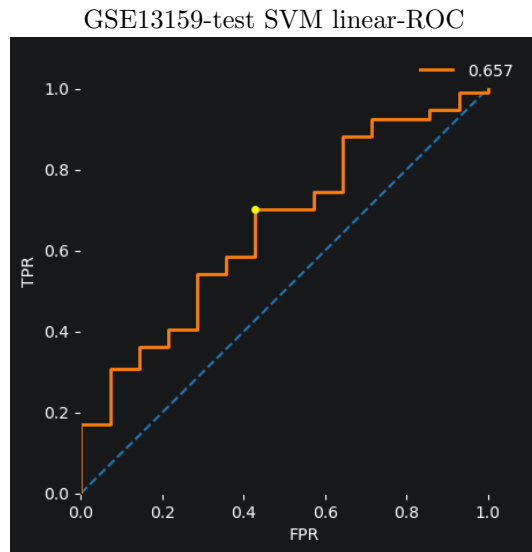
I found that 3 models (Logit, SVM-linear, SVM-linear-weighted) had an ROC-AUC > 0.6. Of course, this might not be the best performance, but considering that I approximated the RPM for the test data with a shift-scale-vector, working on noisy biological, older technology (microarray) and without batch correction, I would argue that this is the right direction to go. Those models could be a classifiers that actually learned biological signals and not just experimental artifacts ...

- Logit

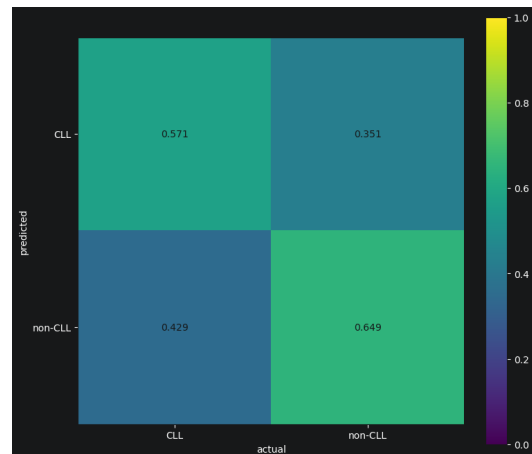


- The CM reveals that the model predicts non-CLL class okish, but performance undesirably, predicting the CLL classes, what is for an diagnostic classifier not appropriate

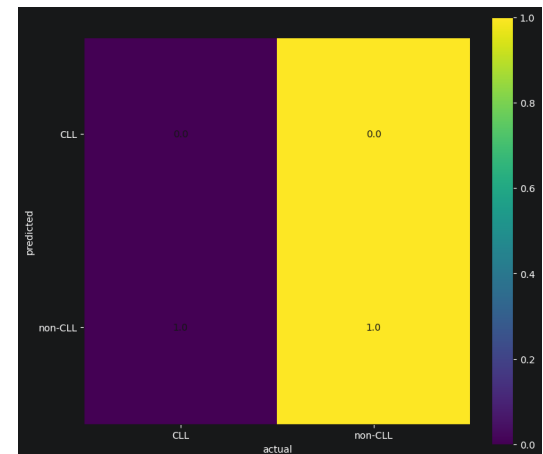
- SVM linear



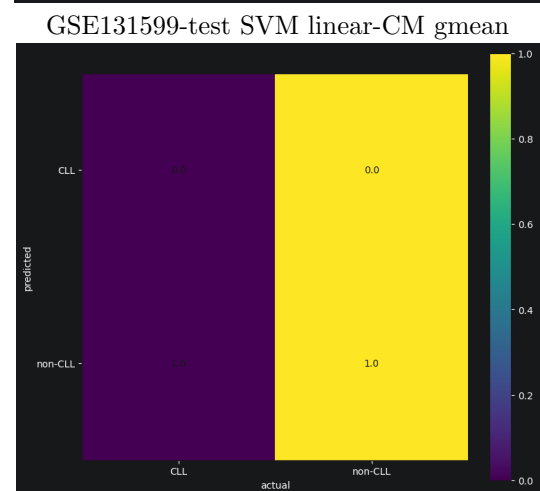
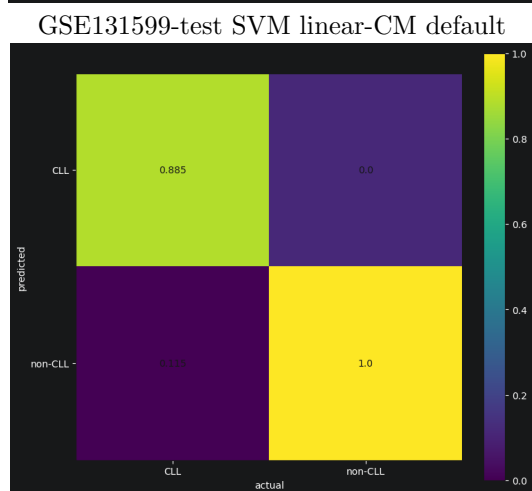
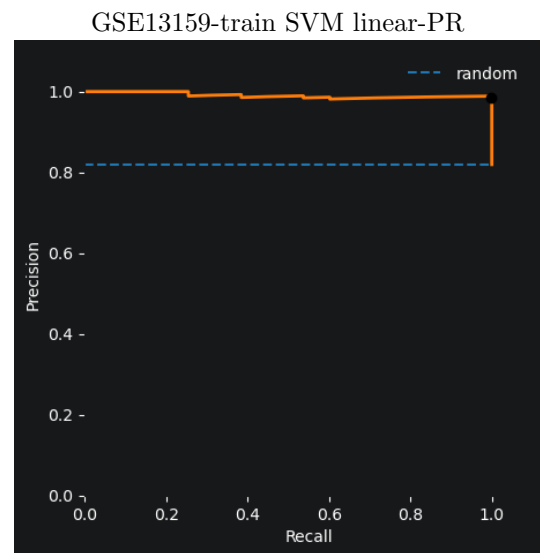
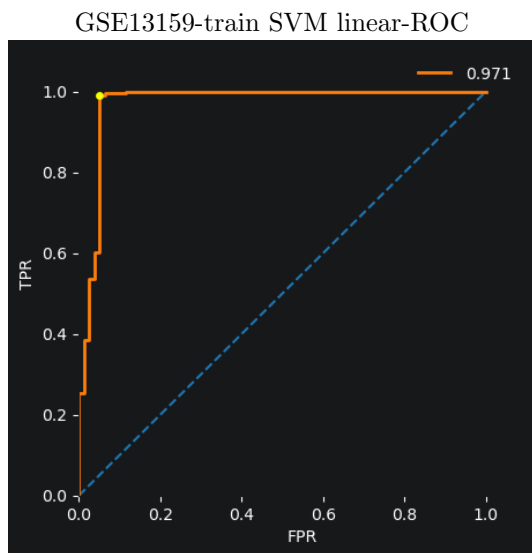
GSE131599-train SVM linear-CM Youden's J statistic



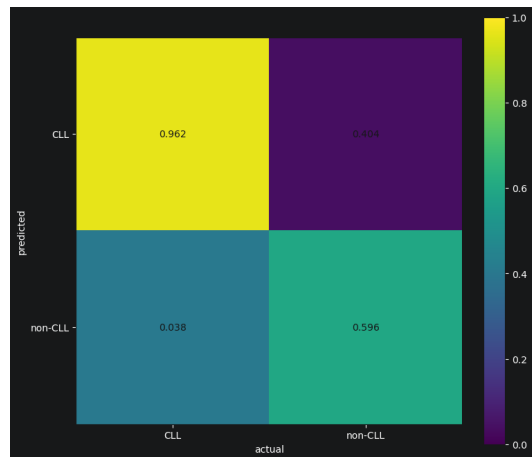
GSE131599-train SVM linear-CM F1-Score



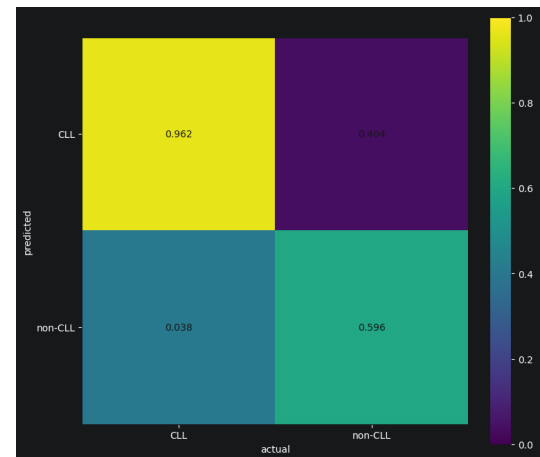
- The default threshold yield a very good non-CLL classifier, but doesn't perform well on the CLL classification
- Two threshold yields a kind of good direction, with 57.1% CLL correct and 64.9% non-CLL correct
- But i would sanity check the train ROC and the CM for the both threshold next



GSE131599-test SVM linear-CM Youden's J statistic

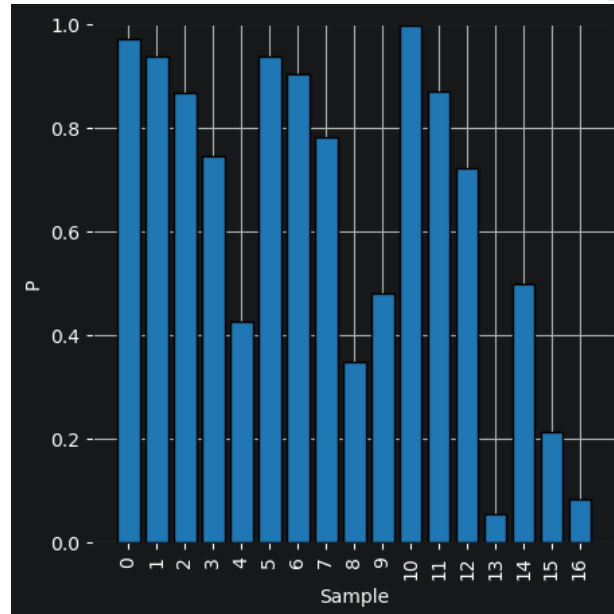


GSE131599-test SVM linear-CM F1-Score



- The classifier performs perfect onto the train data (data seems to be separable with an hyperplane)
- Using the gmean threshold, yield a classifier that only predicts non-cll, so not valid at all
- But using the Youden's statistic threshold, yield a reasonable training classification, and would align with the performance of the GSE13159-test
- So this model and this scaling seems to produce trustworthy predictions
- I would next inspect the GSE16355-test performance,

GSE16455-test SVM-linear Raw-Probabilities for samples

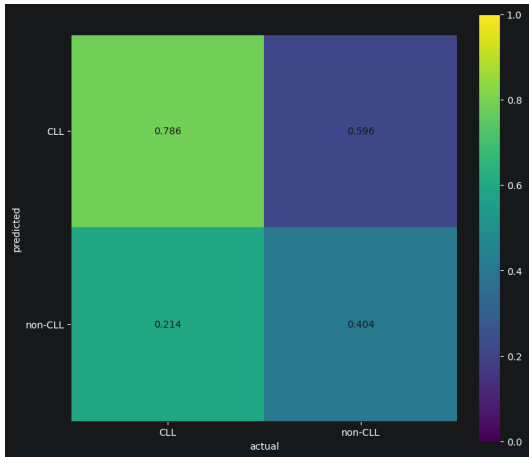
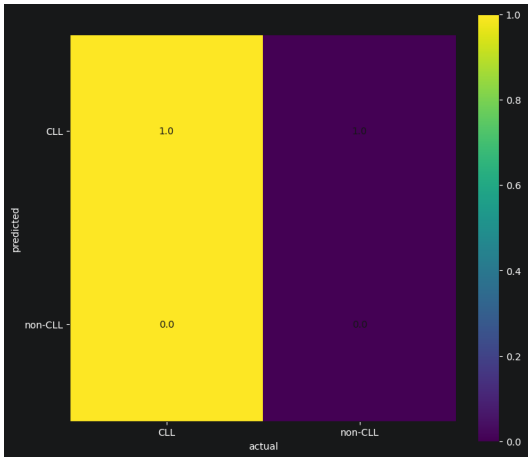
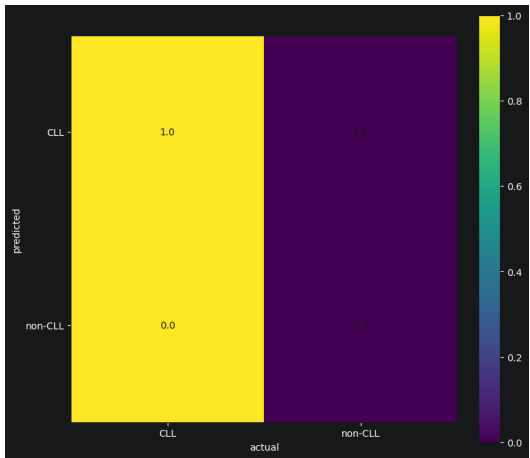
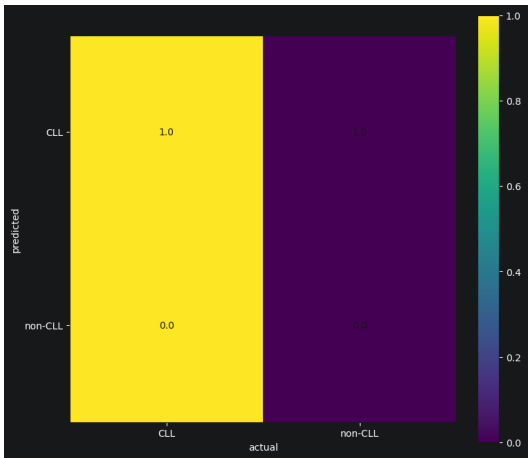
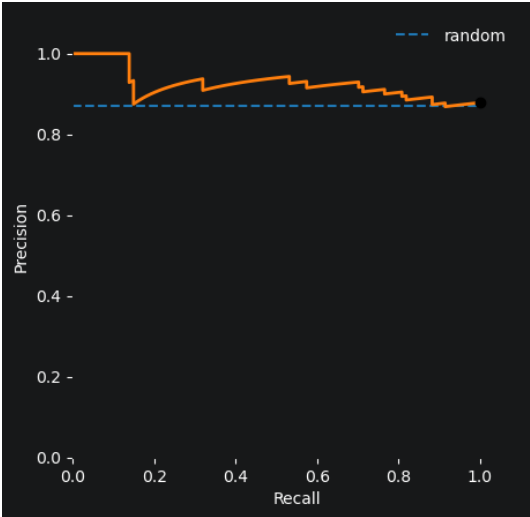
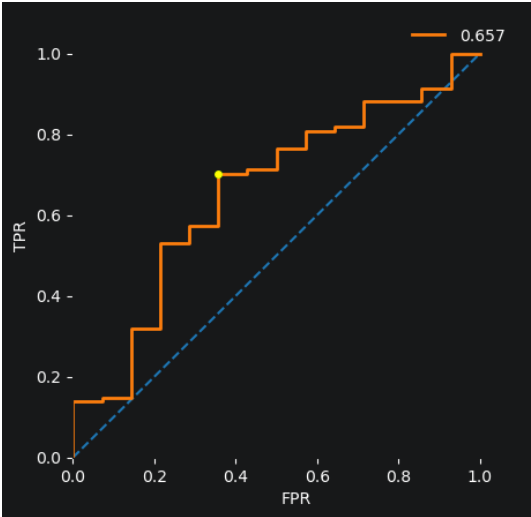


- I investigate the threshold from the Youden's J statistic and found that it is: 0.824
- This would mean, considering this threshold, that 7 / 17 would be classified correctly
- Considering the default threshold of 0.5, i would get 10 / 17 correct, but at that threshold the GSE13159-test performance would predict CLL classes worse.

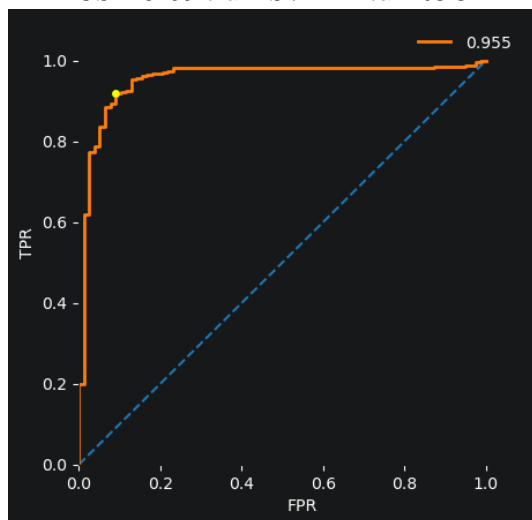
Conclusion

- I would argue that the SVM-linear onto the approximated scaling truly learns biological signal from this inherently noisy data
- I can truly see that the shift-scaled approximation of the RMA seems to work and performs better than just a standard or robust scaling

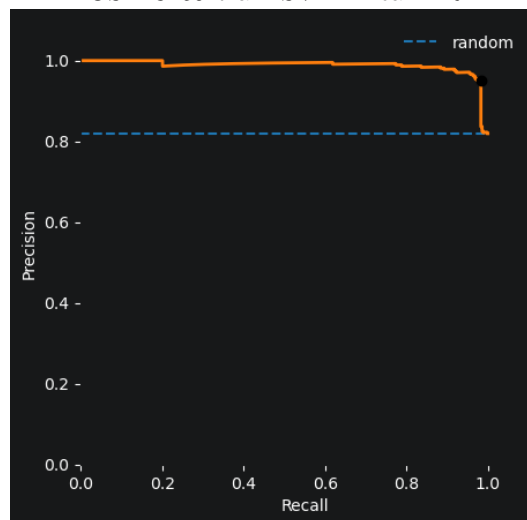
- SVM linear weighted



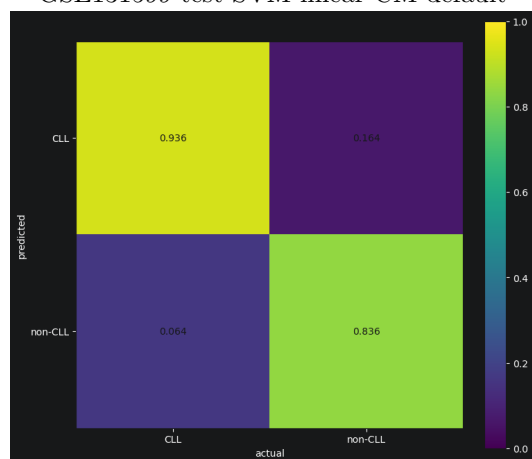
GSE13159-train SVM linear-ROC



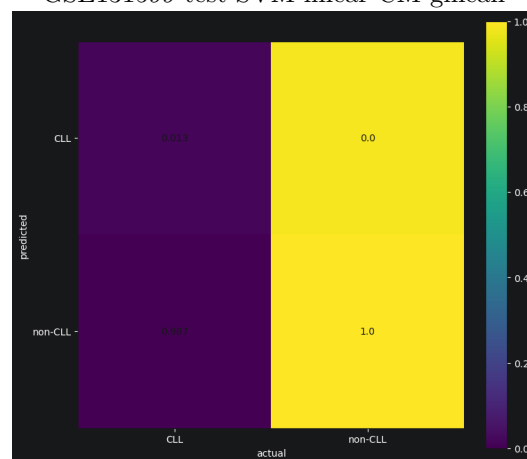
GSE13159-train SVM linear-PR



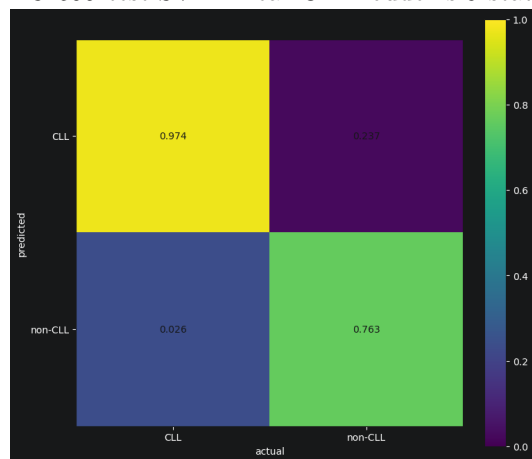
GSE131599-test SVM linear-CM default



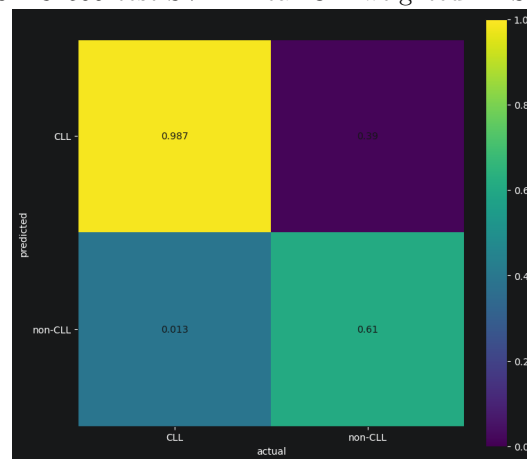
GSE131599-test SVM linear-CM gmean



GSE131599-test SVM linear-CM Youden's J statistic



GSE131599-test SVM linear-CM-weighted F1-Score



GSE16455-test SVM-linear Raw-Probabilities for samples

